

DATA SCIENCE WITH PYTHON PROJECT REPORT

(Project Semester January-May 2026)

Olympic Athlete Performance Analysis and Medal Prediction using Machine Learning

Submitted by

Satvik Dalal

Registration No. 12515479

Programme and Section- M.Tech Data Science & Analytics (325SD)

Course Code- INT557



Under the Guidance of Dr. Karan Bajaj

School of Computer Science and Engineering

Lovely Professional University, Phagwara

DECLARATION

I, Satvik Dalal, student of M.Tech Data Science & Analytics under School of Computer Science and Engineering at Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date: 20-05-2026

Signature

Registration No. 12515479

Satvik Dalal

Lovely Professional University, Phagwara

CERTIFICATE

This is to certify Satvik Dalal, bearing Registration no. 12515479 has completed Data Science with Python (INT557) project titled, **“Olympic Athlete Performance Analysis and Medal Prediction using Machine Learning”** under my guidance and supervision. To the best of my knowledge, the present work is the result of his original development, effort and study.

Signature and Name of the Supervisor: Dr. Karan Bajaj

Designation of the Supervisor: Professor

School of Computer Science and Engineering

Lovely Professional University

Phagwara, Punjab.

Date: 20-05-2026

ACKNOWLEDGEMENT

The successful completion of this project would not have been possible without the support, guidance, and encouragement of many individuals. We feel truly privileged to have received their assistance throughout the journey of this work.

I would like to express my sincere gratitude to **Dr. Karan Bajaj, School of Computer Science and Engineering, Lovely Professional University**, for her invaluable guidance, insightful feedback, and constructive suggestions. Her constant support and encouragement at every stage greatly enhanced the quality of this project. Her mentorship helped me build a strong foundation of knowledge and complete this work effectively.

I am also grateful to all the faculty members of **Lovely Professional University** for their continuous support, motivation, and technical guidance. Their expertise and willingness to help played a significant role in shaping the successful outcome of this project.

TABLE OF CONTENTS

DECLARATION.....	2
CERTIFICATE	3
ACKNOWLEDGEMENT	4
1. Abstract.....	7
2. Introduction	8
3. Problem Statement.....	9
4. Objectives of the Study	10
5. Dataset Description.....	11
5.1 Data Sources.....	11
5.2 Dataset Size and Features	11
5.3 Description of Variables	11
6. Data Preprocessing	13
6.1 Handling Missing Values.....	13
6.2 Data Cleaning	13
6.3 Feature Selection.....	13
6.4 Data Transformation (Encoding, Scaling)	14
7. Exploratory Data Analysis (EDA).....	15
7.1 Distribution of Athletes	15
7.2 Medal Distribution	15
7.3 Performance by Gender.....	15
7.4 Sport-wise Analysis	15
7.5 Age vs Medal Outcome	15
7.6 Height and Weight Analysis.....	16
7.7 Trend Over Years.....	16
7.8 Correlation Analysis	16
8. Data Visualization.....	17
8.1 Participation Over Years	17
8.2 Gender Distribution	17
8.3 Age Distribution of Athletes.....	18
8.4 Medal Count by Gender	19
8.5 Top 10 Sports by Participation	19
8.6 Top 10 Countries by Participation.....	20

8.7 Correlation Heatmap.....	21
8.8 Pairplot Analysis	22
9. Model Building	23
9.1 Decision Tree Classifier	23
9.2 Random Forest Classifier	23
9.3 K-Nearest Neighbors (KNN)	24
9.4 Naive Bayes (GaussianNB)	24
9.5 Logistic Regression	24
9.6 Model Comparison	25
9.7 Performance Analysis	26
9.8 Key Insights	26
9.9 Best Model Selection	26
9.10 Justification	26
10. Insights.....	27
10.1 Key Findings from Data	27
10.2 Behavioral Patterns Observed.....	27
10.3 Overall Insight	27
11. Limitations.....	28
11.1 Data Limitations	28
11.2 Model Constraints.....	28
11.3 Overall Limitation	28
12. Future Scope	29
12.1 Model Improvements	29
12.2 Data Enhancements.....	29
12.3 Feature Engineering Enhancements	29
12.4 Practical Applications.....	29
12.5 Research Extensions	29
12.6 Overall Scope.....	30
13. Conclusion.....	31
14. References	32
14.1 Kaggle Dataset.....	32
14.2 Libraries Used.....	32

1. Abstract

This project focuses on the analysis of Olympic athlete performance and the prediction of medal outcomes using machine learning techniques. The primary objective is to identify the key factors influencing an athlete's likelihood of winning a medal and to build predictive models that can accurately classify whether an athlete will win a medal or not.

To achieve this, a comprehensive dataset containing information on athletes, including demographic attributes (such as age, sex), event details, and historical performance, was utilized. The data underwent thorough preprocessing, including handling missing values, encoding categorical variables, feature selection, and scaling. Various supervised machine learning algorithms, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbours (KNN), and Naive Bayes, were implemented and evaluated.

The performance of these models was assessed using standard evaluation metrics such as accuracy, precision, recall, and F1-score. Comparative analysis was conducted to determine the most effective model for medal prediction. The results indicate that ensemble-based methods, particularly Random Forest, achieved higher predictive performance due to their ability to capture complex patterns in the data.

Overall, the project demonstrates the effectiveness of machine learning in sports analytics and highlights how data-driven approaches can be used to predict athletic success. The findings can assist coaches, analysts, and sports organizations in making informed decisions regarding training strategies and athlete selection.

2. Introduction

In recent years, the field of sports analytics has gained significant attention due to the increasing availability of large-scale data and advancements in machine learning techniques. The Olympic Games, being one of the most prestigious international sporting events, provide a rich source of historical data on athlete performance, participation, and outcomes. Analyzing such data enables researchers and analysts to uncover patterns, trends, and key factors that contribute to athletic success.

Traditionally, performance evaluation in sports has relied on subjective assessments, experience-based judgments, and limited statistical analysis. However, with the emergence of data-driven approaches, it has become possible to perform more objective and accurate analysis. Machine learning techniques allow for the identification of hidden relationships within complex datasets, enabling more reliable predictions of outcomes such as medal wins. This shift from intuition-based to data-driven decision-making has significantly enhanced the effectiveness of performance analysis in sports.

The importance of data-driven analysis lies in its ability to support evidence-based decision-making. By leveraging historical data, sports organizations, coaches, and analysts can better understand the factors that influence performance, optimize training strategies, and improve athlete selection processes. Furthermore, predictive models can help in forecasting outcomes, thereby providing a competitive advantage in highly competitive environments like the Olympics.

The motivation behind this project is to explore how machine learning can be applied to predict Olympic medal outcomes based on athlete and event-related features. By building and evaluating multiple classification models, the project aims to identify the most influential factors contributing to success and determine the most effective predictive approach. This study not only contributes to the domain of sports analytics but also demonstrates the practical application of data science techniques in solving real-world problems.

3. Problem Statement

The Olympic Games involve participation from thousands of athletes competing across diverse sports and events. Despite the availability of extensive historical data, predicting whether an athlete will win a medal remains a complex task due to the influence of multiple factors such as age, gender, type of sport, event category, and prior performance. The challenge lies in identifying the most relevant features and leveraging them effectively to make accurate predictions.

The core problem addressed in this project is to develop a machine learning-based classification system that can predict whether an athlete will win a medal (i.e., Gold, Silver, or Bronze) or not. This problem is formulated as a binary classification task, where the target variable indicates the presence or absence of a medal. The objective is to train models that can learn patterns from historical Olympic data and generalize well to unseen data.

The need for such a predictive system arises from the growing importance of data-driven decision-making in sports. Accurate prediction of medal outcomes can assist coaches, analysts, and sports organizations in evaluating athlete potential, optimizing training strategies, and making informed selections for competitions. Additionally, understanding the key factors influencing medal success can provide valuable insights into performance dynamics across different sports and demographics.

Therefore, this project aims to address the problem by applying and comparing multiple machine learning algorithms to determine the most effective approach for predicting medal outcomes, while also analyzing the significance of various input features in influencing the results.

4. Objectives of the Study

The primary aim of this study is to analyze Olympic athlete data and develop a predictive framework using machine learning techniques to determine the likelihood of winning a medal. The study is designed with the following specific objectives:

1. To analyze athlete performance data Examine the dataset to understand the distribution, patterns, and relationships among various features such as age, gender, sport type, and event category, and how they relate to medal outcomes.
2. To identify key factors influencing medal achievement Determine the most significant variables that contribute to an athlete's success in winning a medal using exploratory data analysis and feature importance techniques.
3. To preprocess and prepare the dataset for modelling Perform data cleaning, handle missing values, encode categorical variables, and apply feature scaling to ensure the dataset is suitable for machine learning algorithms.
4. To build and implement multiple machine learning models Develop various classification models, including Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors (KNN), and Naive Bayes, to predict medal outcomes.
5. To evaluate and compare model performance Assess the performance of each model using evaluation metrics such as accuracy, precision, recall, and F1-score, and identify the most effective model.
6. To analyze model predictions and interpret results Examine how well the models generalize to unseen data and interpret the results to gain insights into athlete performance and prediction reliability.
7. To provide insights for data-driven decision-making in sports Highlight how the findings can be used by coaches, analysts, and sports organizations to improve training strategies, athlete selection, and performance optimization.

5. Dataset Description

This section provides a detailed description of the dataset used in this study. The dataset is sourced from Kaggle and contains comprehensive historical records of Olympic athletes and their performance over more than a century. It serves as the primary foundation for analysis and model development in this project.

5.1 Data Sources

The dataset used in this project is the “120 Years of Olympic History: Athletes and Results” dataset on Kaggle. It is a publicly available dataset compiled from historical Olympic records and scraped from the Sports Reference website.

<https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>

This dataset covers the modern Olympic Games from Athens 1896 to Rio 2016, including both Summer and Winter Olympics.

The dataset is widely used in data science and analytics projects due to its rich historical coverage, structured format, and diverse attributes related to athletes and events. It enables researchers to analyze long-term trends in sports participation, performance, and medal distribution.

5.2 Dataset Size and Features

The primary dataset file, `athlete_events.csv`, contains detailed records of athlete participation in Olympic events.

- Number of Records: 271,116 rows
- Number of Features: 15 columns
- Data Representation: Each row represents an individual athlete competing in a specific Olympic event

The dataset includes both numerical and categorical features, making it suitable for machine learning tasks.

- Numerical Features: Age, Height, Weight, Year
- Categorical Features: Sex, Team, NOC, Season, City, Sport, Event, Medal

The target variable used in this project is derived from the *Medal* column:

- Medal → Converted into binary classification
 - 1 = Medal (Gold/Silver/Bronze)
 - 0 = No Medal

This transformation simplifies the problem into a classification task for predictive modeling.

5.3 Description of Variables

The dataset contains the following key variables:

- ID: Unique identifier assigned to each athlete
- Name: Athlete’s name (used only for reference, not for modeling)
- Sex: Gender of the athlete (Male/Female)

- Age: Age of the athlete during the event
- Height: Height of the athlete in centimeters
- Weight: Weight of the athlete in kilograms
- Team: Country or team represented by the athlete
- NOC: National Olympic Committee code (3-letter country code)
- Games: Combination of year and season (e.g., 2008 Summer)
- Year: Year of the Olympic Games
- Season: Type of Olympics (Summer or Winter)
- City: Host city of the Olympic Games
- Sport: Category of sport (e.g., Athletics, Swimming)
- Event: Specific competition within a sport
- Medal: Outcome of the event (Gold, Silver, Bronze, or No Medal)

6. Data Preprocessing

Data preprocessing is a crucial step in any machine learning project, as the quality of input data directly impacts the performance of predictive models. The Olympic dataset contains missing values, categorical variables, and irrelevant features that must be properly handled before model training. This section outlines the preprocessing techniques applied to prepare the dataset for analysis and modeling.

6.1 Handling Missing Values

The dataset contains missing values, particularly in the Age, Height, Weight, and Medal columns. Appropriate strategies were applied to handle these missing entries:

- Age, Height, Weight: Missing values in these numerical columns were handled using statistical imputation techniques such as replacing them with the mean or median of the respective feature. This approach helps maintain the overall distribution of the data without introducing bias.
- Medal Column: Missing values in the *Medal* column indicate that the athlete did not win any medal. Therefore, these missing values were replaced with “No Medal”, and later converted into a binary format (0 = No Medal, 1 = Medal).
- Rows with excessive or irrelevant missing information were removed where necessary to ensure data quality.

6.2 Data Cleaning

Data cleaning was performed to remove inconsistencies and improve the overall quality of the dataset:

- Removal of Duplicates: Duplicate records, if any, were identified and removed to avoid redundancy.
- Dropping Irrelevant Columns: Columns such as Name and ID, which do not contribute to prediction, were excluded from the modeling process.
- Standardization of Data Formats: Ensured consistent formatting for categorical variables such as *Sex*, *Season*, and *Medal*.
- Outlier Handling: Extreme values in numerical features like Age, Height, and Weight were examined and treated where necessary to prevent distortion in model training.

6.3 Feature Selection

Feature selection was performed to identify the most relevant variables for predicting medal outcomes. Including unnecessary features can lead to overfitting and reduced model performance.

- Relevant features selected include: Sex, Age, Height, Weight, Team/NOC, Year, Season, Sport, Event
- Irrelevant or non-informative features such as Name and ID were excluded.
- Correlation analysis and domain knowledge were used to determine the importance of features.

This step helps in reducing dimensionality and improving model efficiency.

Feature engineering involves creating new variables or modifying existing ones to improve model performance.

- Medal Indicator (Target Variable): The *Medal* column was transformed into a binary variable:
 - 1 → Athlete won a medal (Gold, Silver, Bronze)
 - 0 → Athlete did not win a medal
- Derived Features: Additional meaningful representations were created where applicable, such as grouping or simplifying categories (e.g., combining similar events or sports if required).
- Unlike datasets involving behavioral metrics (e.g., stress score), this dataset focuses on performance-based features, so engineering was centered around outcome representation and categorical simplification.

6.4 Data Transformation (Encoding, Scaling)

Machine learning models require numerical input; therefore, categorical variables were transformed appropriately:

- Encoding Categorical Variables:
 - Label Encoding or One-Hot Encoding was applied to categorical features such as *Sex*, *Season*, *Sport*, *Event*, and *NOC*.
 - One-Hot Encoding was preferred for nominal variables to avoid introducing ordinal relationships.
- Feature Scaling:
 - Numerical features such as Age, Height, and Weight were scaled using techniques like Standardization (StandardScaler) to ensure that all features contribute equally to the model.
 - This step is particularly important for distance-based algorithms like KNN.
- Train-Test Split:
The dataset was divided into training and testing sets to evaluate model performance on unseen data.

7. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a critical step in understanding the structure, patterns, and relationships within the dataset before applying machine learning models. In this study, EDA was performed to gain insights into athlete demographics, participation trends, and factors influencing medal outcomes. Various statistical summaries and visualizations were used to analyze the data.

7.1 Distribution of Athletes

The dataset contains a large number of athlete entries across multiple Olympic years. Analysis of the distribution shows:

- A higher proportion of male athletes compared to female athletes, especially in earlier Olympic years.
- Increasing participation of female athletes in recent years, indicating improved gender representation.
- The age distribution reveals that most athletes fall within the range of 20 to 30 years, which is considered the peak performance age in many sports.

7.2 Medal Distribution

The medal distribution provides insights into the imbalance in the dataset:

- A significant majority of entries correspond to “No Medal”, indicating a class imbalance problem.
- Among medal winners, the distribution of Gold, Silver, and Bronze is relatively balanced.
- This imbalance highlights the need for careful model evaluation and possibly techniques such as resampling or class weighting.

7.3 Performance by Gender

Analysis based on gender shows:

- Male athletes have historically won more medals, primarily due to higher participation rates.
- Female athletes, however, show competitive performance levels in recent Olympic events.
- The gap between male and female participation and performance has decreased over time.

7.4 Sport-wise Analysis

Different sports exhibit varying patterns of participation and medal distribution:

- Sports such as Athletics, Swimming, and Gymnastics have the highest number of participants and medal events.
- Certain sports are more physically demanding and show stronger dependence on attributes like height and weight.
- Team sports and individual sports show different trends in medal distribution.

7.5 Age vs Medal Outcome

- Athletes within the optimal age range (20–30 years) have a higher probability of winning medals.
- Very young or older athletes tend to have lower chances of medal achievement, although exceptions exist depending on the sport.

7.6 Height and Weight Analysis

- Physical attributes such as height and weight play a significant role in certain sports.
- For example, taller athletes may have an advantage in sports like basketball or volleyball, while lower body weight may benefit endurance-based sports.
- However, these factors alone are not sufficient predictors and must be considered along with other variables.

7.7 Trend Over Years

- There is a noticeable increase in the number of participating athletes over time, reflecting the global growth of the Olympic Games.
- Improvements in training, nutrition, and sports science have contributed to better performance and competitiveness.
- Medal distribution trends also indicate increasing diversity among participating countries.

7.8 Correlation Analysis

- Correlation analysis between numerical variables (Age, Height, Weight) shows moderate relationships.
- No single feature strongly determines medal outcomes, indicating that prediction requires a combination of multiple features.
- This justifies the use of machine learning models capable of capturing complex relationships.

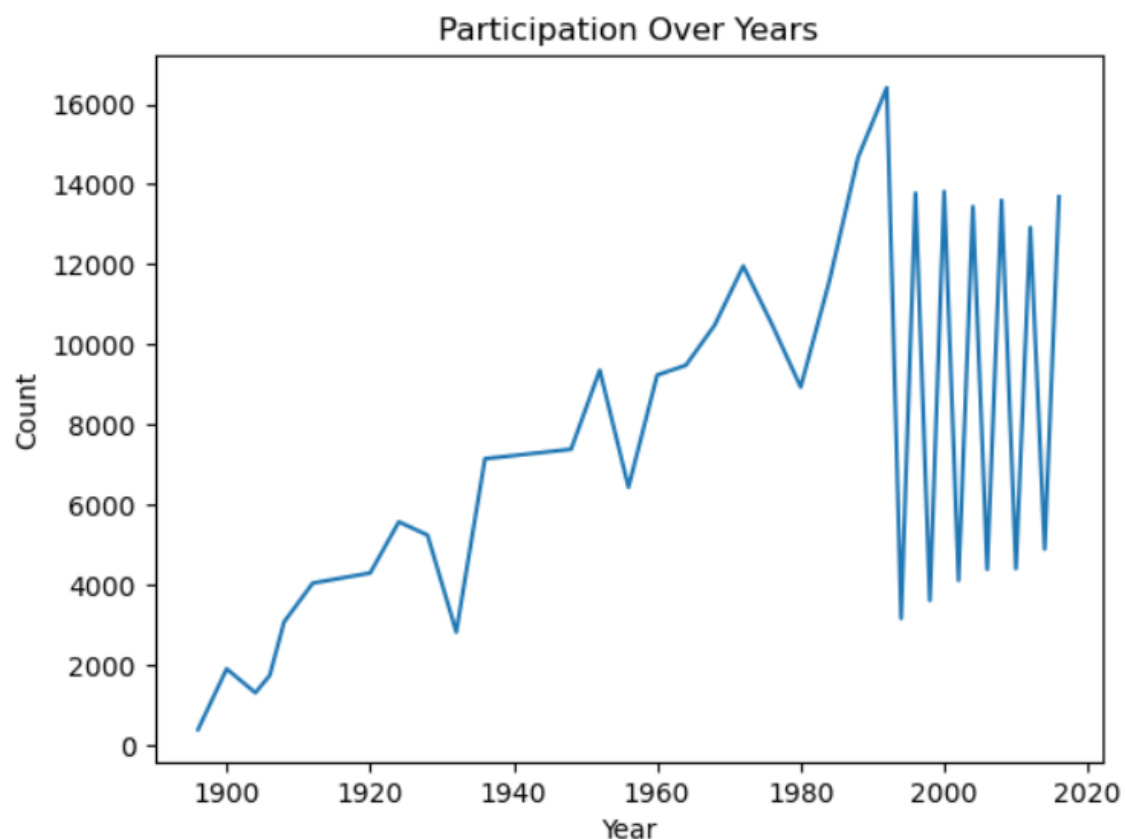
8. Data Visualization

Data visualization techniques were applied to better understand patterns, trends, and relationships within the Olympic dataset. The following visual analyses were performed based on the dataset:

8.1 Participation Over Years

This visualization illustrates how athlete participation in the Olympic Games has evolved over time.

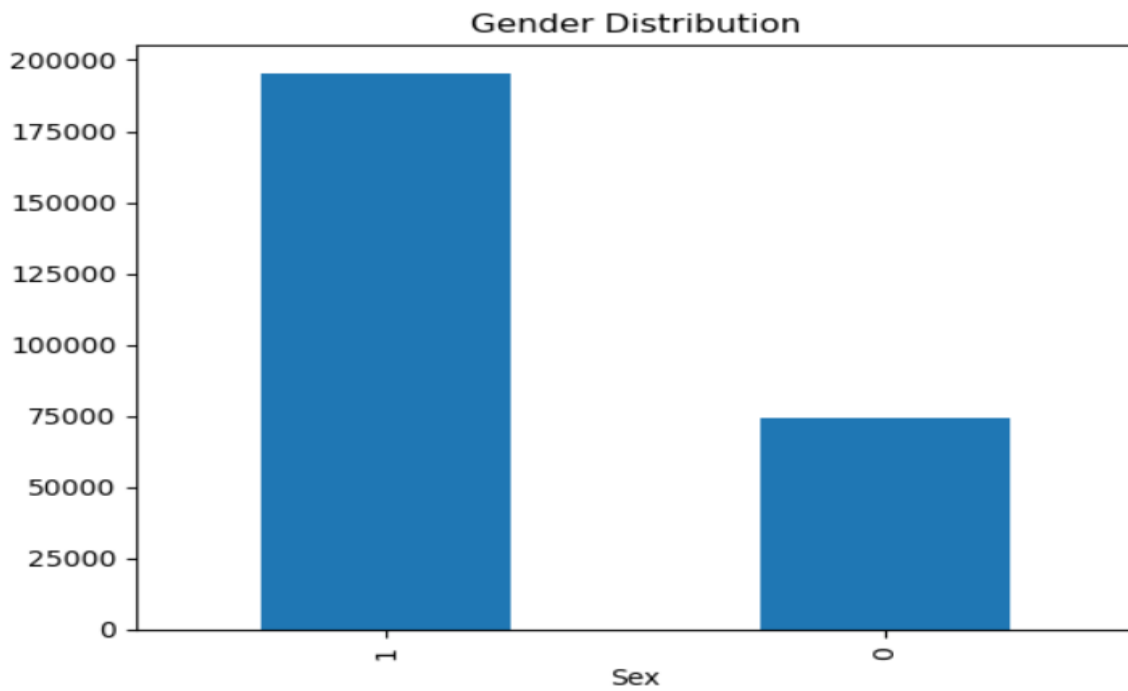
- There is a steady increase in participation from the early 1900s to recent years.
- A noticeable rise occurs after the mid-20th century, reflecting the expansion and globalization of the Olympic Games.
- Some fluctuations are observed due to historical events such as World Wars and economic disruptions.
- In recent years, a zig-zag pattern appears because Summer and Winter Olympics occur alternately, causing variation in participation counts.



8.2 Gender Distribution

This bar chart represents the distribution of athletes based on gender.

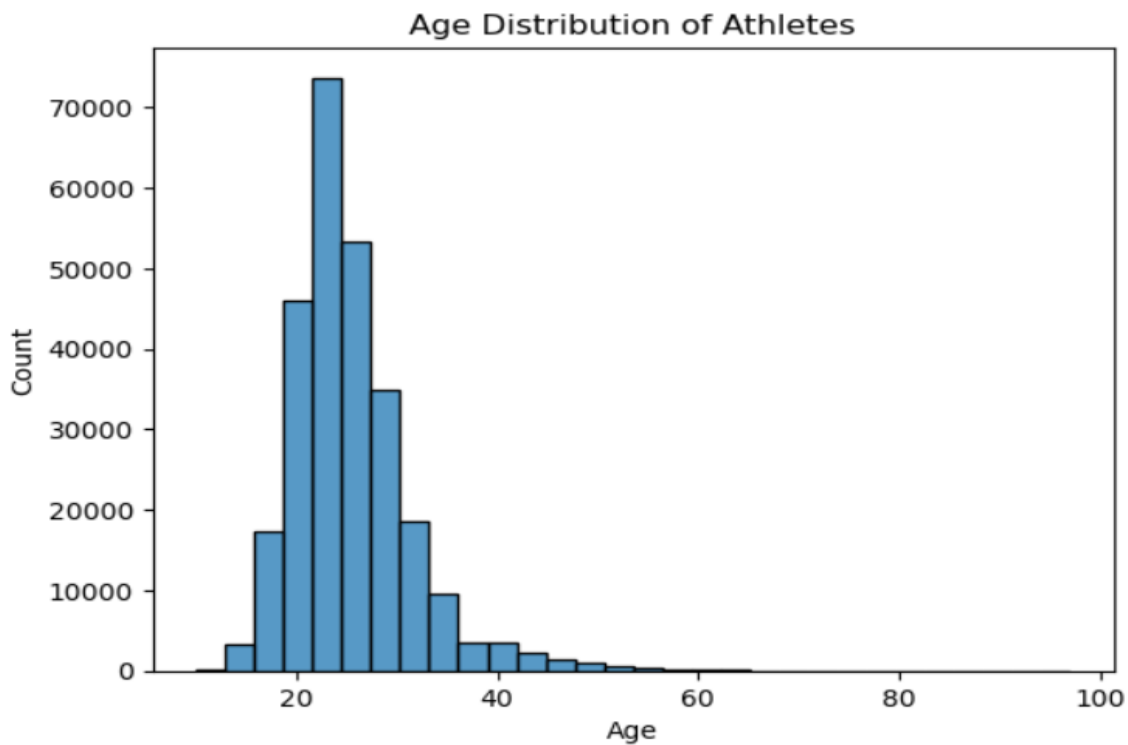
- The dataset shows a significantly higher number of male athletes compared to female athletes.
- This indicates a historical gender imbalance in Olympic participation.
- However, female participation has gradually increased over time, especially in modern Olympic Games.



8.3 Age Distribution of Athletes

This histogram shows the distribution of athletes' ages.

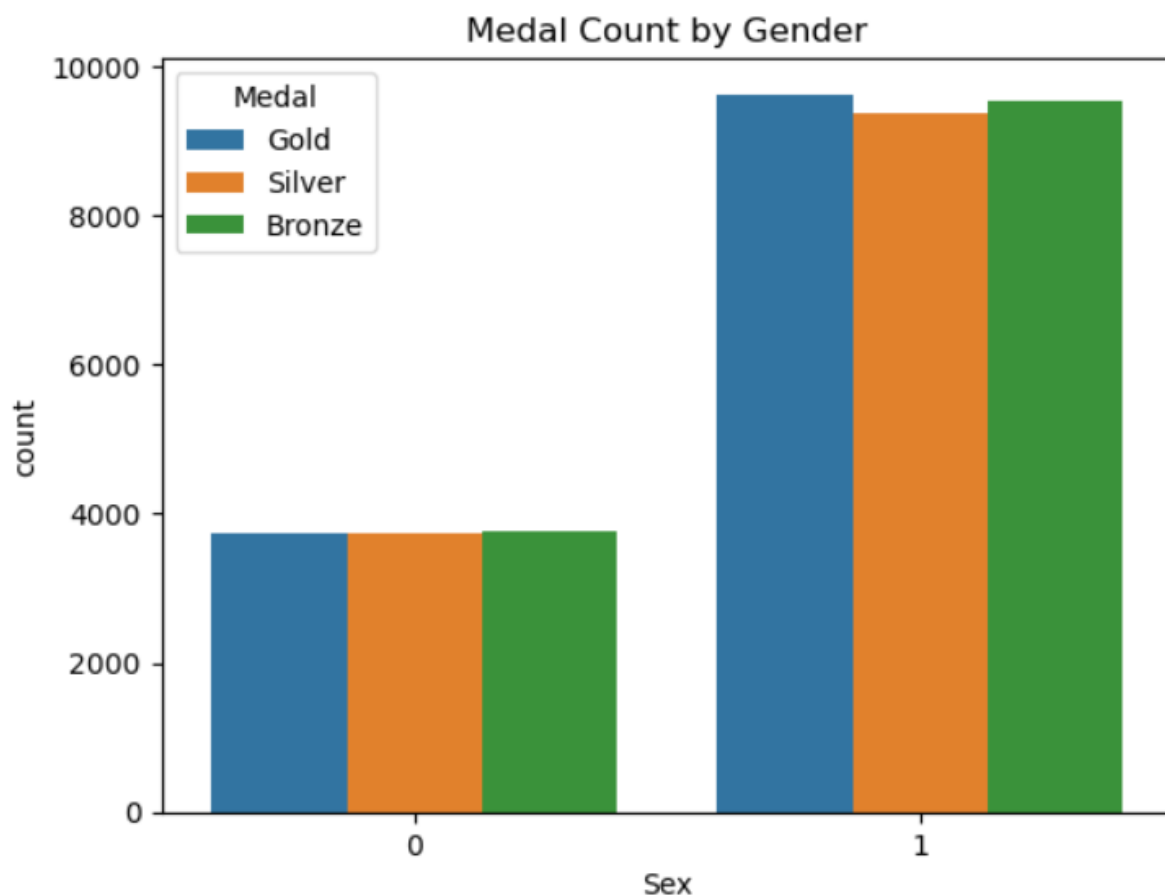
- Most athletes fall within the 20 to 30 years age range.
- The peak occurs around the mid-20s, indicating the most competitive age group.
- There are fewer athletes below 18 and above 35 years.
- A small number of athletes extend into higher age ranges, depending on the sport.



8.4 Medal Count by Gender

This visualization compares the distribution of Gold, Silver, and Bronze medals across genders.

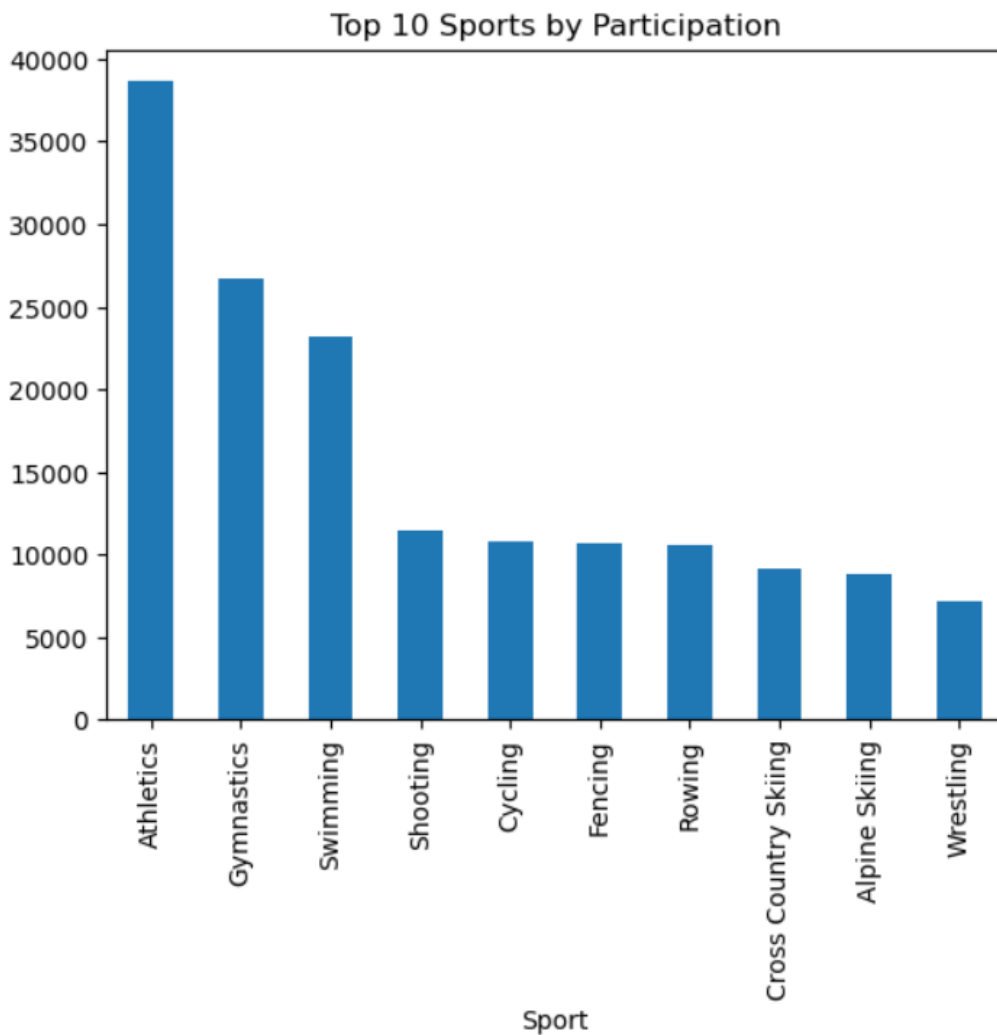
- Male athletes have won more medals overall, mainly due to higher participation.
- The distribution of Gold, Silver, and Bronze medals is relatively balanced within each gender.
- Female athletes have fewer total medals, but their performance has improved significantly in recent years.



8.5 Top 10 Sports by Participation

This bar chart displays the sports with the highest number of athlete participations.

- Athletics has the highest participation among all sports.
- Other highly participated sports include Gymnastics, Swimming, and Shooting.
- These sports often include multiple events, contributing to higher participation numbers.
- Sports like Wrestling and Alpine Skiing have comparatively lower participation among the top 10.



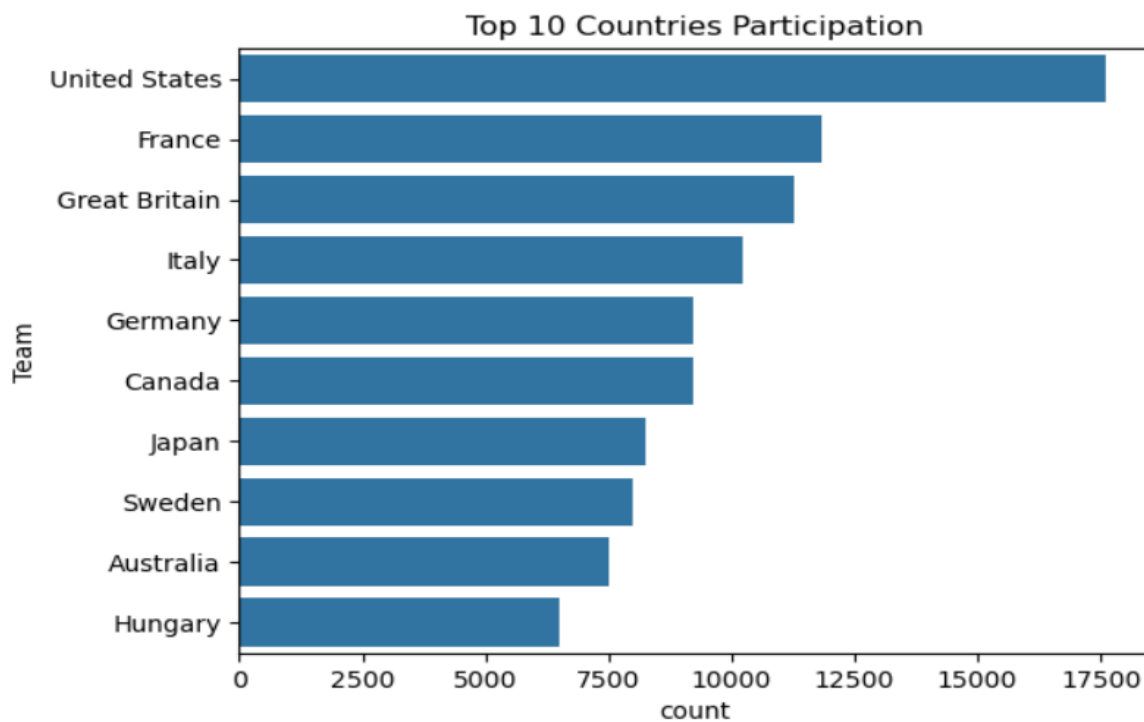
8.6 Top 10 Countries by Participation

This horizontal bar chart shows countries with the highest athlete participation.

- The United States has the highest participation among all countries.
- Other major contributors include France, Great Britain, Italy, and Germany.
- These countries have consistently participated across multiple Olympic years.

Insights:

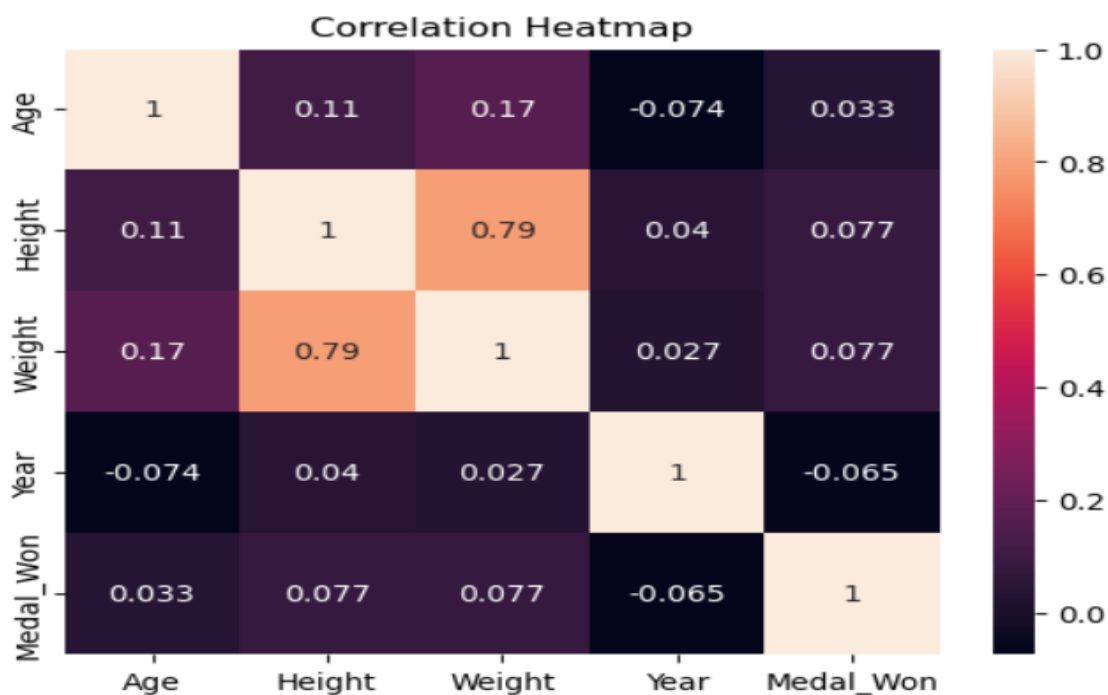
- Countries with strong sports infrastructure and funding tend to have higher participation.
- Developed nations dominate due to better training facilities and support systems.
- Historical involvement also contributes to higher participation levels.



8.7 Correlation Heatmap

This heatmap represents the correlation between numerical features.

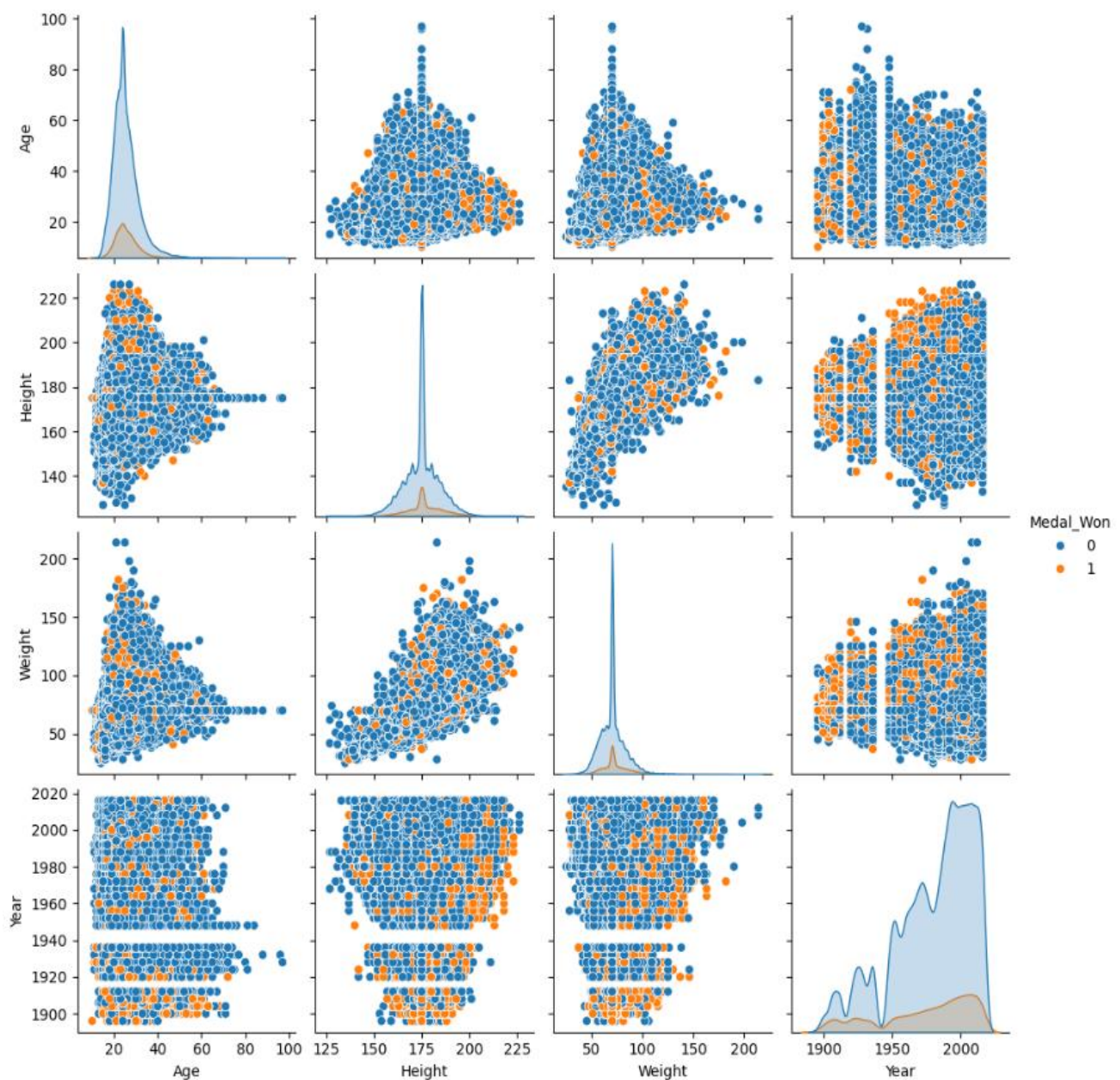
- Height and Weight show a strong positive correlation, indicating that taller athletes tend to weigh more.
- Age and Weight/Height show weak correlations.
- Year has very low correlation with physical attributes.
- The target variable (Medal_Won) has very weak correlation with individual features, suggesting that no single factor determines medal success.



8.8 Pairplot Analysis

The pairplot visualizes relationships between multiple numerical features and their connection to medal outcomes.

- It shows the distribution and interaction between Age, Height, Weight, and Year.
- Medal winners and non-winners are spread across similar ranges, indicating overlapping patterns.
- Some clustering patterns can be observed, but clear separation is not present.
- This suggests that multiple features combined are required for accurate prediction rather than relying on a single variable.



9. Model Building

In this study, multiple machine learning models were developed to predict whether an athlete will win a medal. The problem was formulated as a binary classification task, where the target variable (*Medal_Won*) indicates whether an athlete won a medal (1) or not (0).

The input features used for model training include: Sex, Age, Height, Weight, and Year.

The dataset was divided into training and testing sets (80:20 split) using a random state to ensure reproducibility. Model performance was evaluated using Accuracy, Precision, Recall, F1-Score, Confusion Matrix, and ROC-AUC Curve.

9.1 Decision Tree Classifier

The Decision Tree model was implemented as a baseline model to understand decision-making patterns.

- The model achieved an accuracy of approximately 72%.
- Precision was relatively low, indicating incorrect positive predictions.
- Recall was moderate, meaning some medal-winning athletes were correctly identified.

Confusion Matrix Insights:

- High number of True Negatives (correctly predicted non-medal athletes)
- Moderate True Positives
- Presence of both False Positives and False Negatives

ROC-AUC Score: ~ 0.59

- Slightly better than random prediction

Feature Importance:

- Year was the most important feature
- Followed by Weight
- Other features (Age, Height, Sex) had lower importance

Interpretation:

The Decision Tree model captures basic patterns but struggles with generalization and produces moderate performance.

9.2 Random Forest Classifier

Random Forest was used to improve performance by combining multiple decision trees.

- Accuracy improved to approximately 77%
- Precision and recall showed better balance compared to Decision Tree

Confusion Matrix Insights:

- Increased True Negatives
- Reduced False Positives compared to Decision Tree
- Still some False Negatives present

ROC-AUC Score: ~ 0.64

- Indicates moderate predictive performance

Feature Importance:

- Features such as Year, Height, Weight, and Age contributed almost equally
- Sex had the least importance

Interpretation:

Random Forest provides better generalization and handles feature interactions effectively, making it more reliable than a single Decision Tree.

9.3 K-Nearest Neighbors (KNN)

The KNN algorithm classifies data based on similarity with neighboring data points.

- Accuracy was relatively high ($\sim 83\%$)
- However, precision and recall were imbalanced

Confusion Matrix Insights:

- Very high True Negatives
- Very low True Positives
- High number of False Negatives, meaning many medal winners were missed

Interpretation:

Although KNN shows high accuracy, it is misleading due to class imbalance. The model is biased towards predicting non-medal outcomes.

9.4 Naive Bayes (GaussianNB)

Naive Bayes was applied as a probabilistic model assuming feature independence.

- Accuracy was around 85%
- However, performance on the minority class (medal winners) was poor

Confusion Matrix Insights:

- Extremely high True Negatives
- Very few True Positives
- High number of False Negatives

Interpretation:

The model performs well for majority class prediction but fails to correctly identify medal winners, making it unsuitable for this problem.

9.5 Logistic Regression

Logistic Regression was used as a linear classification model.

- Accuracy was approximately 58–60%
- Recall was relatively high (~ 0.55)

- Precision was low (~0.18)

Confusion Matrix Insights:

- Increased number of True Positives compared to other models
- High False Positives, reducing precision

ROC-AUC Score: ~ 0.62

- Indicates moderate performance

Interpretation:

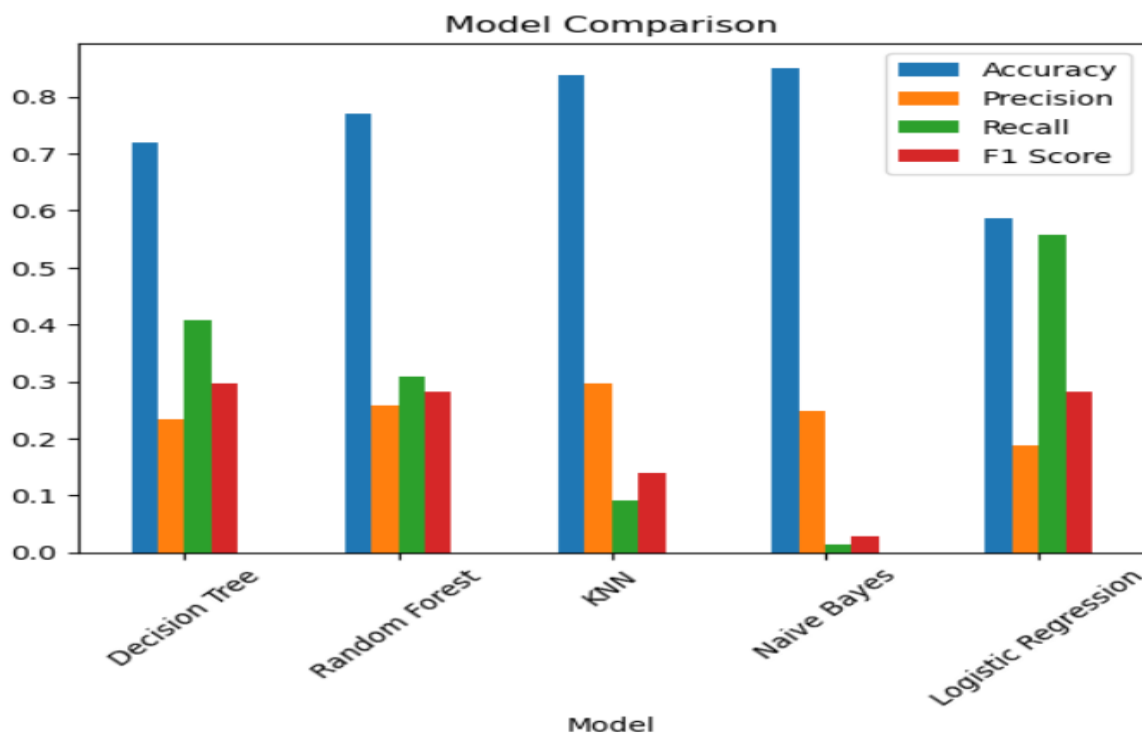
Logistic Regression performs better in identifying medal winners (higher recall), but at the cost of more false alarms (low precision).

9.6 Model Comparison

To evaluate the performance of different machine learning models, a comparative analysis was conducted using key evaluation metrics such as Accuracy, Precision, Recall, and F1 Score. These metrics provide a comprehensive understanding of how well each model performs, especially in the presence of class imbalance.

The results of all models are summarized as follows:

Model	Accuracy	Precision	Recall	F1 Score
Decision Tree	72.0%	23.4%	40.7%	29.7%
Random Forest	77.1%	25.8%	30.9%	28.1%
KNN	83.7%	29.7%	9.1%	13.9%
Naive Bayes	85.6%	24.7%	1.4%	2.7%
Logistic Regression	58.6%	18.8%	55.7%	28.1%



9.7 Performance Analysis

Accuracy

- Naive Bayes and KNN show the highest accuracy.
- However, this is misleading due to class imbalance, as these models tend to predict the majority class (No Medal) more often.

Precision

- KNN achieves the highest precision, indicating better correctness in positive predictions.
- However, this comes at the cost of very low recall.

Recall

- Logistic Regression has the highest recall, making it most effective at identifying medal-winning athletes.
- This is important when detecting true positives is a priority.

F1 Score

- Decision Tree achieves the highest F1 Score, indicating the best balance between precision and recall.
- This makes it more reliable for overall performance comparison.

9.8 Key Insights

- Models such as Naive Bayes and KNN are biased toward the majority class, leading to poor detection of medal winners.
- Logistic Regression focuses on identifying medal winners, but produces a high number of false positives.
- Random Forest provides balanced performance across multiple metrics and shows good generalization.
- Decision Tree achieves the best trade-off between precision and recall, as reflected by the highest F1 Score.

9.9 Best Model Selection

Based on the overall evaluation:

- The Decision Tree Classifier is selected as the best-performing model.
- It achieves the highest F1 Score (~0.297), which is the most appropriate metric for imbalanced datasets.
- It provides a balanced performance in identifying both medal and non-medal athletes.

9.10 Justification

The selection of the Decision Tree model is justified based on the following factors:

- Highest F1 Score, indicating an optimal balance between precision and recall
- Better capability to identify medal-winning athletes compared to most models
- Interpretable structure, allowing easier understanding of decision-making
- Handles non-linear relationships effectively

10. Insights

This section summarizes the key findings and behavioral patterns observed from the data analysis, visualization, and machine learning modeling. These insights provide a deeper understanding of athlete performance and the factors influencing Olympic medal outcomes.

10.1 Key Findings from Data

- **Participation Growth Over Time:** There has been a consistent increase in athlete participation over the years, reflecting the global expansion and popularity of the Olympic Games.
- **Gender Imbalance and Improvement:** Male athletes dominate the dataset, especially in earlier years. However, female participation has significantly increased in recent Olympics, indicating progress toward gender equality.
- **Age as a Performance Factor:** Most successful athletes fall within the 20–30 years age range, suggesting this as the optimal performance window for many sports.
- **Dominance of Certain Sports:** Sports such as Athletics, Swimming, and Gymnastics have the highest participation and medal counts, mainly due to the large number of events.
- **Country-wise Performance Trends:** Countries like the United States, Great Britain, and France consistently show high participation, indicating strong sports infrastructure and long-term involvement.

10.2 Behavioral Patterns Observed

- **Class Imbalance in Medal Outcomes:** A large proportion of athletes do not win medals, creating a highly imbalanced dataset. This significantly impacts model performance and evaluation.
- **Weak Individual Feature Influence:** No single feature (Age, Height, Weight, etc.) strongly determines medal success. Instead, performance depends on a combination of multiple factors.
- **Physical Attributes Matter but Are Not Decisive:** Height and weight show relevance in certain sports but are not universally strong predictors of medal outcomes.
- **Temporal Influence (Year):** The *Year* feature shows importance in models, suggesting that performance trends evolve over time due to advancements in training, technology, and competition levels.
- **Model Behavior Patterns:**
 - Some models (Naive Bayes, KNN) tend to favor the majority class (No Medal)
 - Logistic Regression emphasizes detecting medal winners (high recall)
 - Decision Tree and Random Forest provide more balanced predictions

10.3 Overall Insight

The analysis reveals that Olympic performance is influenced by a complex interaction of demographic, physical, and temporal factors, rather than any single variable. Machine learning models are capable of capturing these patterns, but their effectiveness depends on handling class imbalance and selecting appropriate evaluation metrics.

These insights highlight the importance of adopting a data-driven approach in sports analytics to better understand performance trends and improve predictive accuracy.

11. Limitations

Despite the successful implementation of machine learning models and insightful analysis, this study has several limitations related to both the dataset and the modeling approach. These limitations may affect the accuracy and generalizability of the results.

11.1 Data Limitations

- **Class Imbalance:** The dataset contains a significantly higher number of non-medal instances compared to medal-winning cases. This imbalance affects model performance, leading to biased predictions toward the majority class.
- **Missing Values and Imputation:** Important features such as Age, Height, and Weight contain missing values that were handled using statistical imputation. This may reduce data accuracy and introduce bias.
- **Limited Feature Set:** The dataset lacks important variables such as training intensity, athlete experience, coaching quality, injury history, and nutrition, which are crucial factors influencing performance.
- **Event-Specific Variations Not Fully Captured:** Different sports have unique physical and skill requirements, but the model uses generalized features, which may reduce prediction accuracy across diverse events.
- **Temporal Bias:** The inclusion of data spanning over 120 years introduces variations due to changes in rules, technology, and competition levels, which are not explicitly modeled.

11.2 Model Constraints

- **Simplified Feature Representation:** Only a limited number of features (Sex, Age, Height, Weight, Year) were used for modeling, which restricts the predictive capability of the models.
- **Handling of Class Imbalance:** Although techniques like class weighting were used, more advanced methods (e.g., SMOTE, oversampling, undersampling) were not fully explored.
- **Model Bias Toward Majority Class:** Some models (e.g., Naive Bayes, KNN) tend to favor predicting the majority class (No Medal), resulting in poor detection of medal winners.
- **Limited Hyperparameter Tuning:** Models were implemented with basic configurations. Advanced tuning techniques such as Grid Search or Random Search could improve performance.
- **Moderate Predictive Performance:** The ROC-AUC scores and evaluation metrics indicate that the models achieve only moderate performance, suggesting that the problem is inherently complex.
- **Lack of Deep Learning or Advanced Models:** More advanced approaches such as neural networks or ensemble boosting methods (e.g., XGBoost) were not applied.

11.3 Overall Limitation

The primary limitation of this study lies in the complex and multifactorial nature of athletic performance, which cannot be fully captured using limited structured data. While the models provide useful insights, they should not be considered definitive predictors of real-world outcomes.

12. Future Scope

This project provides a foundational approach to predicting Olympic medal outcomes using machine learning. However, there are several opportunities for improvement and extension that can enhance the accuracy, robustness, and applicability of the study.

12.1 Model Improvements

- **Advanced Algorithms:** Future work can incorporate more powerful models such as XGBoost, LightGBM, and Neural Networks, which are capable of capturing complex non-linear relationships more effectively.
- **Hyperparameter Tuning:** Techniques such as Grid Search and Random Search can be applied to optimize model parameters and improve performance.
- **Handling Class Imbalance:** Advanced resampling techniques such as SMOTE (Synthetic Minority Oversampling Technique), undersampling, or ensemble balancing methods can be used to improve prediction of medal-winning athletes.
- **Cross-Validation:** Implementing k-fold cross-validation can provide more reliable and generalized model evaluation.

12.2 Data Enhancements

- **Inclusion of Additional Features:** Incorporating more detailed attributes such as athlete experience, training history, injury records, coaching quality, and diet can significantly improve prediction accuracy.
- **Event-Specific Modeling:** Developing separate models for different sports or event categories can better capture sport-specific performance factors.
- **Temporal Analysis:** Applying time-series or trend-based modeling can help account for changes in performance patterns over different Olympic years.

12.3 Feature Engineering Enhancements

- **Derived Performance Metrics:** Create new features such as experience level (number of previous participations), age categories, BMI (Body Mass Index), and country performance index.
- **Country-Level Aggregation:** Incorporate historical medal counts or country-wise performance trends to improve predictive insights.

12.4 Practical Applications

- **Athlete Performance Prediction Systems:** The model can be extended into a real-world application to assist coaches and sports organizations in evaluating athlete potential.
- **Sports Analytics Dashboards:** Develop interactive dashboards for visualizing trends, predictions, and performance insights.
- **Decision Support Systems:** Integrate the model into systems that support team selection, training optimization, and strategic planning.

12.5 Research Extensions

- **Deep Learning Approaches:** Explore neural network architectures for handling complex feature interactions.

- Ensemble Learning: Combine multiple models to create hybrid systems for improved prediction accuracy.
- Explainable AI (XAI): Use techniques such as SHAP or LIME to interpret model predictions and understand feature contributions in depth.

12.6 Overall Scope

The future scope of this project lies in enhancing both data quality and model sophistication. By incorporating richer datasets, advanced algorithms, and domain-specific knowledge, the predictive system can be transformed into a more accurate and practical tool for sports analytics and decision-making.

In the future, this system can be expanded to support real-time athlete performance monitoring by integrating wearable sensor data, fitness tracking information, and live competition statistics. Such integration can help coaches and analysts make instant decisions regarding athlete performance and training adjustments. Additionally, deep learning techniques and hybrid ensemble models can further improve prediction accuracy by capturing hidden and complex relationships within the data.

The project can also be extended to predict not only medal outcomes but also the type of medal (Gold, Silver, or Bronze), athlete ranking, and team performance. Integration with cloud-based analytics platforms and interactive dashboards can make the system more accessible for sports organizations and researchers. Furthermore, the use of Explainable AI (XAI) techniques can improve transparency and trust in predictions by clearly identifying the factors influencing athlete success. Overall, this project has strong potential to contribute significantly to the future of intelligent sports analytics and data-driven decision-making.

Additionally, future research can focus on integrating psychological and environmental factors such as athlete motivation, mental health, climate conditions, audience pressure, and competition experience, which often influence athletic performance but are difficult to capture using traditional datasets. Including these factors can provide a more holistic understanding of athlete success and improve the reliability of predictions.

Another important extension of this work could involve the use of big data technologies and distributed computing frameworks to handle large-scale sports datasets efficiently. As Olympic and sports-related data continue to grow rapidly, technologies such as Apache Spark and Hadoop can support faster processing and large-scale analytics. This would allow researchers and organizations to analyze millions of performance records in real time.

From an academic perspective, this project creates opportunities for interdisciplinary research combining data science, sports science, artificial intelligence, psychology, and healthcare analytics. Educational institutions and sports academies can also use such predictive systems for talent identification and athlete development programs. In the long term, intelligent sports prediction systems may become an essential part of strategic planning, performance optimization, and competitive analysis in professional sports industries worldwide.

13. Conclusion

This project aimed to analyze Olympic athlete data and develop machine learning models to predict whether an athlete will win a medal. By utilizing the “120 Years of Olympic History: Athletes and Results” dataset, a comprehensive analysis was conducted, including data preprocessing, exploratory data analysis, visualization, and model building.

The study revealed several important insights regarding athlete performance. It was observed that factors such as age, gender, physical attributes, and temporal trends play a role in influencing outcomes, although no single feature independently determines medal success. The analysis also highlighted challenges such as class imbalance, which significantly impacts model performance and evaluation.

Multiple machine learning models, including Decision Tree, Random Forest, K-Nearest Neighbors (KNN), Naive Bayes, and Logistic Regression, were implemented and compared. Each model demonstrated different strengths and limitations. While models like Naive Bayes and KNN achieved higher accuracy, they were biased toward the majority class and failed to effectively identify medal winners. On the other hand, Logistic Regression showed higher recall, making it more effective in detecting medal-winning athletes, albeit with lower precision.

Among all models, the Decision Tree Classifier emerged as the best-performing model based on the highest F1 Score, which provides a balanced measure of precision and recall. It demonstrated a more reliable trade-off between correctly identifying medal winners and minimizing false predictions. Random Forest also showed strong performance and better generalization, making it a competitive alternative.

In conclusion, the project successfully demonstrates the application of machine learning in sports analytics for predicting Olympic medal outcomes. Although the models achieved moderate performance due to the complexity of the problem, the study highlights the potential of data-driven approaches in understanding athletic performance and supporting decision-making in sports. Future improvements in data quality, feature engineering, and advanced modeling techniques can further enhance prediction accuracy and practical applicability.

14. References

This section lists the data sources and libraries used in the development of this project.

14.1 Kaggle Dataset

- 120 Years of Olympic History: Athletes and Results Kaggle. Available at: <https://www.kaggle.com/datasets/heesoo37/120-years-of-olympic-history-athletes-and-results>

This dataset provides historical records of Olympic athletes, including demographic details, event participation, and medal outcomes from 1896 to 2016.

14.2 Libraries Used

The following Python libraries were used for data analysis, visualization, and machine learning:

- NumPy
Used for numerical operations and handling arrays.
- Pandas
Used for data manipulation, cleaning, and preprocessing.
- Matplotlib
Used for creating basic visualizations such as histograms and bar charts.
- Seaborn
Used for advanced visualizations such as heatmaps and pair plots.
- Scikit-learn
Used for implementing machine learning models including Logistic Regression, Decision Tree, Random Forest, KNN, and Naive Bayes.